

Every Memory Needs a WHY

Outcome-Grounded Memory Consolidation for Continual AI Agents

WHY Research · why.com · Ockams Inc.

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Abstract

Memory is becoming a standard component of AI agents, and any memory with finite capacity is a selection policy. Many write policies decide what to keep from signals available when an experience happens: recency, frequency, similarity, a model’s own importance rating, or user approval. We argue that consolidation should instead target *counterfactual learning value*, the improvement in future decisions caused by keeping an experience, and that because this value cannot be observed at write time, consolidation should wait for consequences. We propose *consequence-weighted consolidation*. Episodes stay in a short-lived buffer until their outcomes resolve. Each receives a WHY score combining surprise, consequence, causal confidence, transferability, and uncertainty reduction, discounted when the outcome arrived through a manipulable channel, and episodes are rescored every night so a later outcome can retroactively promote an earlier experience. We test the hypothesis in a controlled simulation (60 days, 6,000 decisions, 13 policies, 5 scenarios, 20 seeds) using a Bayesian agent whose durable memory is capped at 300 episodes and whose counterfactual memory value is computable. In a realistic scenario with late, missing, noisy, and shock-confounded outcomes and approval bias, consequence-weighted consolidation reached a decision quality of 0.974 (0 = random choice, 1 = optimal), against 0.946 for the best other capacity-limited policy, captured 65% of a counterfactual oracle’s headroom over random retention, and beat the recency, similarity, recurrence, self-rated-importance, and user-preference baselines on every seed. Under a self-report poisoning campaign it lost 0.004 in decision quality while self-rated importance lost 0.331. The support is conditional. With immediate, trustworthy outcomes the advantage over random retention shrinks to +0.006. Ablations credit waiting for outcomes, discounting manipulable channels, and detecting confounded days rather than surprise or stakes, and the surprise factor is what makes the method worse calibrated. When poisoned outcomes pass verification, the method admits more poison than random retention. Unbounded memory beats it in every stationary scenario but collapses when the world changes (0.618 versus 0.965). We release code and preregister an LLM-agent study with explicit falsification criteria.

1 Introduction

Agent memory has become a product category: longer context windows, vector stores, memory operating systems, and assistants that promise to remember everything [7, 26, 40, 43]. But capacity is always finite, whether it is measured in context tokens, retrieval precision, or the compute and interference cost of writing experience into weights. So every memory system is a selection policy. Something decides what is written, what is kept, and what is allowed to shape future behavior.

Many published and deployed write policies make that decision with signals available at write time: how recent an experience is, how often similar experiences recur, how similar it is to current queries, how important the model itself rates it (Generative Agents asks the model to score poignancy from 1 to 10 [27]), how emotionally salient it is (a recent sleep-inspired design weights emotional valence directly [32]), or whether the user approved. A growing line of work uses task outcomes instead. Memory-R1 learns memory operations with outcome rewards [41]; ReasoningBank distills strategies from self-judged successes and failures [25]; TraceRetain scores retention with success and downstream-utility features [29]; Memory Worth

tracks how often a memory co-occurs with success [34]. These systems are typically evaluated where the environment reports success immediately and reliably, such as household tasks or question answering.

Deployed agents rarely get that. Consequences arrive late: a forecast resolves in weeks, and a code change fails in production days after it shipped. They are partial, because most interactions never report back. They are unreliable, because people misremember and misreport. They are confounded, because an outage fails every action taken that day. And they are adversarial, because memory injection attacks can write misleading experiences through ordinary interaction [6, 8]. The most available signal, approval, is also the most dangerous. OpenAI’s postmortem of an April 2025 GPT-4o update attributes part of a sycophancy regression to an added reward signal built from thumbs-up and thumbs-down data, noting that “user feedback in particular can sometimes favor more agreeable responses” [24].

We argue that the target of consolidation should be counterfactual learning value: an experience deserves to change a system’s future behavior if keeping it measurably improves future decisions. Approval tells you how something felt. Outcomes tell you what happened. Because learning value cannot be observed when an experience occurs, a consolidation policy should wait for consequences, weigh them by how attributable and trustworthy they are, and revise its judgment when a later outcome reveals that an earlier experience mattered.

Hypothesis. Given limited consolidation capacity, selecting episodes according to verified downstream learning value produces better future decision-making than FIFO memory, similarity-based retrieval, recurrence-based consolidation, self-rated salience, or user-preference signals.

Contributions.

1. A formulation of consolidation as budgeted selection that targets counterfactual memory value, with a greedy counterfactual oracle as a reference (Section 3).
2. *Consequence-weighted consolidation*: a WHY score, deferral until outcomes resolve with nightly rescoring (retroactive tagging), discounting of manipulable outcome channels, and redundancy control through sequential posterior updates, run as an offline dream cycle (Section 4).
3. A controlled environment that independently varies outcome delay, verification, exogenous confounds, approval bias, poisoning, and nonstationarity, and a comparison of 13 policies across 5 scenarios and 20 seeds, with ablations and sweeps. We report the conditions where the method loses (Sections 5–6).
4. A preregistered protocol for LLM agents, with falsification criteria (Section 7).

Scope. Experiment 1 uses a Bayesian logistic agent, not a language model. It isolates the selection principle in a setting where the true value of a memory can be computed. It does not show that the principle transfers to LLM agents; Section 7 specifies the experiment that would.

2 Background and related work

Selective consolidation in brains. Biological memory is selective. Sleep appears to triage memories rather than strengthen all of them equally [36], and information people expect to need later benefits more from sleep [39]. Emotional arousal modulates consolidation through the amygdala [21], and reward modulates hippocampal replay [2, 35]. Tagging theories propose that weak traces can be stabilized by later, stronger events [11, 23], and behavioral studies report retroactive memory enhancement for items related to later emotional learning [10] or reward [28]; the reward effect, however, did not replicate consistently across four experiments [37]. Mattar and Daw formalize replay prioritization as the expected value of a memory access, the product of *gain* (how much a backup improves the policy) and *need* (how often the relevant situation will recur) [19]. Complementary learning systems theory frames the division of labor between fast episodic

storage and slow consolidation [16, 20]. We use these ideas as design analogies. None of them is evidence for the engineering claim tested here.

Prioritized replay and data valuation. Prioritized experience replay samples transitions in proportion to their temporal-difference error [31]. The value of a single training point can be defined by leave-one-out or influence [15] and generalized with Shapley values [12]; RHO-LOSS selects points that are learnable, worth learning, and not yet learned [22]; continual learning has studied which samples to keep in small replay buffers [1, 5]. Our ValueOfMemory is a leave-one-out value and our WHY score is a cheap proxy for it. What differs is the data: labels arrive late, sometimes never, and through channels of different reliability.

Memory for LLM agents. Generative Agents retrieve by recency, importance, and relevance [27]; MemGPT manages memory tiers [26]; MemoryBank applies a forgetting curve [43]; A-MEM and Mem0 organize and update memories [7, 40]; Reflexion stores verbal self-reflections [33]. Outcome-aware approaches include ReasoningBank [25], Memory-R1 [41], Memory-R2, which reduces credit-assignment bias for memory operations with rerollouts from a shared memory state [42], TraceRetain [29], and Memory Worth, whose authors note that it is associational rather than causal [34]. A recent survey lists continual consolidation, causally grounded retrieval, and learned forgetting among the open problems [9].

Consolidation into weights. Sleep-like replay reduces catastrophic forgetting in artificial networks [38]. SEAL trains a model to generate its own fine-tuning data using downstream performance as reward [44]; Titans gates test-time memorization by a surprise signal [3]; “Language Models Need Sleep” consolidates knowledge on an architectural schedule with a dreaming phase for self-improvement [4]; sleep-time compute spends offline compute anticipating future queries [17]. World-model agents “dream” in a different sense, by learning in imagination [14].

What is new here. We do not claim to introduce outcome-based memory management, replay prioritization, or data valuation. The contribution is narrower. We treat the outcome channel itself (delay, verification, attribution, manipulability) as a first-class input to consolidation; we make consolidation decisions revisable after the fact; and we test these choices against the write-time heuristics that dominate agent memory, and against unbounded memory, in an environment where every policy can be compared with a counterfactual oracle and where the conditions that should matter can be switched on and off.

3 Problem formulation

An *episode* $e = (x, a, \hat{p}, y, \tau, \kappa, v)$ records the decision context and options x , the action a , the agent’s predicted success $\hat{p}$, the outcome $y \in \{0, 1\}$ observed at resolution time τ (possibly ∞) through channel $\kappa \in \{\text{verified}, \text{self-report}\}$, and the stakes v . The agent keeps a short-lived episodic buffer B_t (the last H days) and a durable memory M_t with $|M_t| \leq C$. Its decision policy π_M is fit on durable memory only. Future decision quality is

$$J(M) = \mathbb{E}_{q \sim \mathcal{D}_{\text{future}}} [v_q p^*(\pi_M(q))], \quad (1)$$

where p^* is the true success probability. The value of keeping an episode is

$$\text{ValueOfMemory}(e \mid M) = J(M \cup \{e\}) - J(M), \quad (2)$$

and nightly consolidation with budget b is

$$S_t = \arg \max_{S \subseteq B_t, |S| \leq b} J(M_t \cup S). \quad (3)$$

Three facts shape the method. Equation 2 is a leave-one-out value, so it is not additive across episodes and greedy selection is a natural approximation. It is unobservable at write time, because it depends on y , τ , κ , p^* , and $\mathcal{D}_{\text{future}}$. And an episode can have negative value: a corrupted, confounded, or poisoned outcome makes the policy worse. The *counterfactual oracle* in our experiments solves Equation 3 greedily with access to p^* and to each episode’s eventual label; it is a reference point, not a true upper bound.

4 Consequence-weighted consolidation

4.1 The WHY score

For an episode whose outcome has resolved, we compute

$$\text{WHY}(e) = \prod_{f \in \{S, C, K, T, U\}} [\varepsilon + (1 - \varepsilon) f(e)] \times (1 - \lambda_M M(e)), \quad (4)$$

with every factor in $[0, 1]$, a floor $\varepsilon = 0.05$ so no single factor can zero an episode, and $\lambda_M = 0.7$.

- **Surprise** $S(e) = |y - \hat{p}(e)|$ under the *current* model, recomputed every night.
- **Consequence** $C(e) = \min(v_e/\bar{v}, 3)/3$, stakes relative to the buffer mean.
- **Causal confidence** $K(e)$ asks whether the outcome is attributable to the decision. Our implementation detects day-level confounds: for each resolution day it computes the standardized deviation z of all outcomes that resolved that day from the model’s expectation, relative to the previous week’s median, and sets $K = \sigma(-2(|z| - 3))$. A day on which many unrelated decisions fail together is treated as a shock, not as evidence about the decisions.
- **Transferability** $T(e)$ is expected reuse (the “need” term of [19]): mean kernel similarity between the episode and the last two days of tasks, normalized to $[0, 1]$.
- **Uncertainty reduction** $U(e) = \frac{1}{2} \log(1 + w_e x_e^\top \Sigma x_e)$ with $w_e = \hat{p}(1 - \hat{p})$, the information the episode adds to the posterior, normalized by the night’s maximum.
- **Manipulability** $M(e) = 1$ if the outcome arrived through a channel that an interested party can write (a self-report), and 0 if it was verified.

Surprise and uncertainty reduction overlap but are not redundant: S needs the outcome and measures how wrong the model was; U does not, and measures how much the model can still learn in that region. $S \cdot U$ plays the role of Mattar and Daw’s gain, T of need, C converts to value units, and K and M weigh the reliability of the evidence. We implement the penalty for manipulability as a multiplicative discount, which avoids mixing units, and handle redundancy by sequential selection (below). All constants were fixed before any evaluation run; none were tuned.

4.2 Retroactive tagging

An episode whose outcome has not resolved is not scored as unimportant; it is *deferred*. It stays eligible for as long as it remains in the buffer and is rescored every night. Its own late outcome can therefore promote it, and so can a change in the model that makes an already-resolved but unremarkable episode informative. We also test a propagation variant, loosely analogous to behavioral tagging, in which a surprising, consequential outcome boosts the transferability of similar buffered episodes.

4.3 Redundancy

Selection is greedy. After each pick $e = (x, y)$ with $w = \hat{p}(1 - \hat{p})$, the posterior takes a rank-one Laplace step,

$$\theta \leftarrow \theta + \frac{\Sigma x (y - \hat{p})}{1 + w x^\top \Sigma x}, \quad \Sigma \leftarrow \Sigma - \frac{w \Sigma x x^\top \Sigma}{1 + w x^\top \Sigma x}, \quad (5)$$

Algorithm 1 Dream cycle (one night)

Require: buffer B_t , durable memory M_t , posterior (θ, Σ) , budget b , capacity C

- 1: $E \leftarrow \{e \in B_t : e \text{ not consolidated, } \tau_e \leq t\}$ $\triangleright$ unresolved episodes are deferred, not discarded
 - 2: compute causal confidence per resolution day; compute S, C, K, T, U, M for $e \in E$
 - 3: $S_t \leftarrow \emptyset$
 - 4: **for** $i = 1$ **to** $\min(b, |E|)$ **do**
 - 5: $e^* \leftarrow \arg \max_{e \in E \setminus S_t} \text{WHY}(e)$; $S_t \leftarrow S_t \cup \{e^*\}$
 - 6: rank-one update of (θ, Σ) with e^* ; recompute S and U on $E \setminus S_t$
 - 7: **end for**
 - 8: $M_{t+1} \leftarrow M_t \cup S_t$; evict the oldest engrams while $|M_{t+1}| > C$
 - 9: refit (θ, Σ) on engrams in M_{t+1} whose outcomes have resolved
 - 10: episodes older than H days leave the buffer; those never consolidated are forgotten
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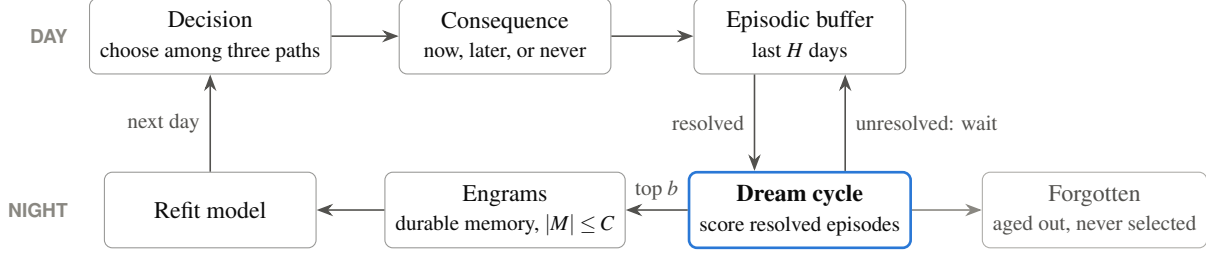


Figure 1: Consequence-weighted consolidation. During the day the agent decides, and consequences resolve on their own schedule. At night the dream cycle scores buffered episodes whose outcomes are known, consolidates at most b of them into capacity-limited durable memory, and lets the rest either wait for their outcomes or age out.

where both updates use the pre-update Σ , and S and U are recomputed for the remaining candidates. Ten near-duplicates of the same lesson stop looking informative once one of them is in memory.

4.4 What forgetting means here

An episode is forgotten when it leaves the buffer without being consolidated. Durable memory evicts its oldest engrams first under every policy, so the policies we compare differ only in what they admit. Figure 1 shows the loop.

5 Experiment 1: controlled simulation

Tasks. Each day the agent makes 100 decisions drawn from eight task families. Every decision offers three options described by twelve features: eight rule features (each family activates four), an agreeableness feature, and three distractors. An option succeeds with probability $\sigma(\theta^{*\top}x + b_f)$, where the rule weights satisfy $|\theta_j^*| \in [0.6, 1.8]$, agreeableness carries a small negative weight (-0.3 ; flattering options are slightly worse), distractors carry none, and $b_f \sim \mathcal{N}(0, 0.3^2)$ is a family offset the agent cannot model. Three held-out families recombine the same rule features and never appear during learning; we use them to measure transfer. About 35% of decisions are near-duplicates of routine prototypes.

Stakes and delays. Each decision has stakes $v \sim \text{LogNormal}(\mu_f, 0.5)$ with three family tiers. Outcomes resolve the same day, after one to six days, or never, and higher-stakes tiers are more often delayed (same-day/delayed/never: 70/10/20%, 50/30/20%, 30/50/20%). Across a run, 20% of outcomes never resolve and 35% of resolvable outcomes arrive late.

Table 1: Consolidation policies. “Outcome” marks policies whose selection uses resolved outcomes.

Policy	Selection signal	Outcome
Random	uniform over buffered episodes	
Random (resolved only)	uniform over episodes whose outcomes have resolved; isolates the value of waiting	✓
Recency (FIFO)	most recent episodes	
Recurrence	episodes whose topic recurred most often in the buffer	
Similarity to recent queries	mean kernel similarity to the last two days of tasks	
Self-rated importance	simulated rater: drama + 0.5 × standardized log stakes + noise, at write time, in the style of poignancy ratings [27]	
User preference (selection)	thumbs-up episodes; trains on outcomes	
User preference (as reward)	same selection; trains on approval instead of outcomes	
Surprise (PER-style)	sampled $\propto (y - \hat{p} + 0.01)^{0.6}$ among resolved episodes [31]	✓
Surprise (greedy top- b)	the b most surprising resolved episodes	✓
WHY	Equation 4, retroactive tagging, redundancy control	✓
Counterfactual oracle	greedy Equation 3 with true p^* and eventual labels	✓
Keep everything	every episode, unbounded capacity	

Channels, shocks, approval, salience. About 30% of outcomes arrive by self-report, which flips the label with probability 0.2; the rest are verified. On shock days (probability 0.07 per day after day 5; 3.6 per run on average) 60% of the outcomes that resolve that day are forced to failure whatever the decision was, so they are real consequences that say nothing about the decision. After each decision the user gives a thumbs-up with probability $\sigma(1.5g + 1.5(2p^* - 1))$, where g is the chosen option’s agreeableness: approval tracks quality but rewards flattery. A drama score (2 with probability 0.15, plus Gaussian noise) marks emotionally salient episodes and is unrelated to learning value.

Scenarios. *Clean*: same-day verified outcomes, no approval bias, no drama, no shocks. *Realistic*: everything above. *Poisoning*: Realistic plus, on days 25–34, 20 extra episodes per day whose self-reported outcomes follow a false rule (success if and only if a distractor is positive), with maximal approval and drama and same-day resolution. *Verified poisoning*: the same false outcomes arrive through the verified channel. *World change*: Realistic, but the two largest rule weights flip sign at night 30. For a given seed all scenarios share the same base task stream, outcomes, and noise (common random numbers).

Agent and memory. The agent is a Bayesian logistic regression (prior $\mathcal{N}(0, 1.5^2 I)$; MAP by damped Newton with Armijo backtracking; Laplace posterior) that acts by Thompson sampling [30]. Durable memory is its only training data: each night it is refit on the engrams whose outcomes have resolved. The buffer holds $H = 8$ days of episodes; each night a policy may consolidate up to $b = 20$ episodes into durable memory of capacity $C = 300$ (5% of a run’s decisions). We chose this agent because it makes $J(M)$ exactly computable, which lets us compare every policy with a counterfactual oracle and trace failures to specific memories.

Policies. Table 1 lists the 13 policies. Two are references rather than competitors: the counterfactual oracle and *keep everything*, which consolidates every episode with unbounded capacity and therefore is not subject to the capacity limit in the hypothesis.

Metrics. *Decision quality* is the stakes-weighted true success of the agent’s greedy choices on a fixed probe of 500 tasks from the training families, normalized so that random choice scores 0 and optimal choice scores 1; unless stated otherwise we report its mean over nights 40–59. *Transfer* is the same quantity on 500 tasks

Table 2: Realistic scenario (delays, missing and noisy outcomes, shocks, approval bias), 20 seeds, mean $\pm$ 95% CI. Decision quality and transfer: nights 40–59, 0 = random choice, 1 = optimal. Headroom: share of the oracle’s advantage over random retention. Rows above the rule are capacity-limited ($C = 300$); keep everything is unbounded.

Policy	Decision quality $\uparrow$	Transfer $\uparrow$	ECE* $\downarrow$	Regret (%) $\downarrow$	Headroom
Random	0.932 $\pm$ 0.014	0.912 $\pm$ 0.016	0.061 $\pm$ 0.007	19.0 $\pm$ 0.9	0%
Random (resolved only)	0.946 $\pm$ 0.010	0.921 $\pm$ 0.017	0.059 $\pm$ 0.006	14.8 $\pm$ 0.8	22%
Recency (FIFO)	0.927 $\pm$ 0.011	0.917 $\pm$ 0.016	0.058 $\pm$ 0.009	19.3 $\pm$ 1.1	<0%
Recurrence	0.870 $\pm$ 0.018	0.822 $\pm$ 0.046	0.048 $\pm$ 0.005	29.8 $\pm$ 1.5	<0%
Similarity to recent queries	0.902 $\pm$ 0.022	0.883 $\pm$ 0.023	0.043 $\pm$ 0.005	25.4 $\pm$ 2.2	<0%
Self-rated importance	0.931 $\pm$ 0.013	0.920 $\pm$ 0.011	0.055 $\pm$ 0.009	19.3 $\pm$ 1.5	<0%
User preference (selection)	0.921 $\pm$ 0.017	0.898 $\pm$ 0.026	0.050 $\pm$ 0.007	20.4 $\pm$ 1.3	<0%
User preference (as reward)	0.370 $\pm$ 0.057	0.347 $\pm$ 0.037	0.386 $\pm$ 0.021	88.1 $\pm$ 2.0	<0%
Surprise (PER-style)	0.935 $\pm$ 0.010	0.913 $\pm$ 0.016	0.115 $\pm$ 0.008	18.9 $\pm$ 1.6	4%
Surprise (greedy top- b)	0.498 $\pm$ 0.022	0.467 $\pm$ 0.024	0.132 $\pm$ 0.008	70.7 $\pm$ 1.4	<0%
WHY	0.974$\pm$0.004	0.969$\pm$0.004	0.114 $\pm$ 0.005	13.3 $\pm$ 1.0	65%
Counterfactual oracle	0.996 $\pm$ 0.001	0.993 $\pm$ 0.002	0.105 $\pm$ 0.021	13.8 $\pm$ 2.4	100%
Keep everything (unbounded)	0.993 $\pm$ 0.002	0.990 $\pm$ 0.003	0.077 $\pm$ 0.006	5.0 $\pm$ 0.4	95%

from the held-out families. *Calibration error* (ECE*) compares the predicted success of each greedy choice with its true success probability in ten bins [13]. *Regret* is the stakes-weighted expected regret of the agent’s actual decisions over all 60 days, as a percentage of a uniformly random policy’s regret. *Oracle headroom captured* is $(q - q_{\text{random}})/(q_{\text{oracle}} - q_{\text{random}})$ for decision quality q . *Poison damage* is decision quality over nights 25–44 minus the same policy’s value in the Realistic scenario on identical base streams.

Protocol. Environment parameters and WHY constants were fixed before evaluation. Development used seeds 1000–1004; every number reported here comes from evaluation seeds 0–19, with common random numbers across policies. We report means with 95% t -intervals across seeds and paired differences. Three protocol deviations occurred (Appendix D). The most consequential was a solver bug found after the first evaluation run; we discarded that run and re-ran every suite.

6 Results

6.1 Main comparison

Table 2 and Figure 2 (left) summarize the Realistic scenario. Consequence-weighted consolidation reached a late decision quality of 0.974 ± 0.004 , above every other capacity-limited policy. The strongest of those was random retention restricted to resolved outcomes (0.946); WHY exceeded it by $+0.028 \pm 0.011$ and was higher on 20 of 20 seeds. Against the baselines named in the hypothesis, the paired differences were $+0.046$ for recency, $+0.072$ for similarity, $+0.104$ for recurrence, $+0.043$ for self-rated importance, and $+0.053$ for user preference, each positive on all 20 seeds. WHY captured 65% of the oracle’s headroom over random retention, compared with 22% for the best competitor, and the gain carried over to held-out task families (transfer 0.969 versus 0.921).

About 33% of WHY’s gain over plain random retention comes from waiting. Restricting random selection to resolved outcomes lifts decision quality from 0.932 to 0.946, because a fifth of all outcomes never resolve and random retention fills about a fifth of its memory with episodes that will never carry a label (Table 3). The rest comes from which resolved episodes the score selects.

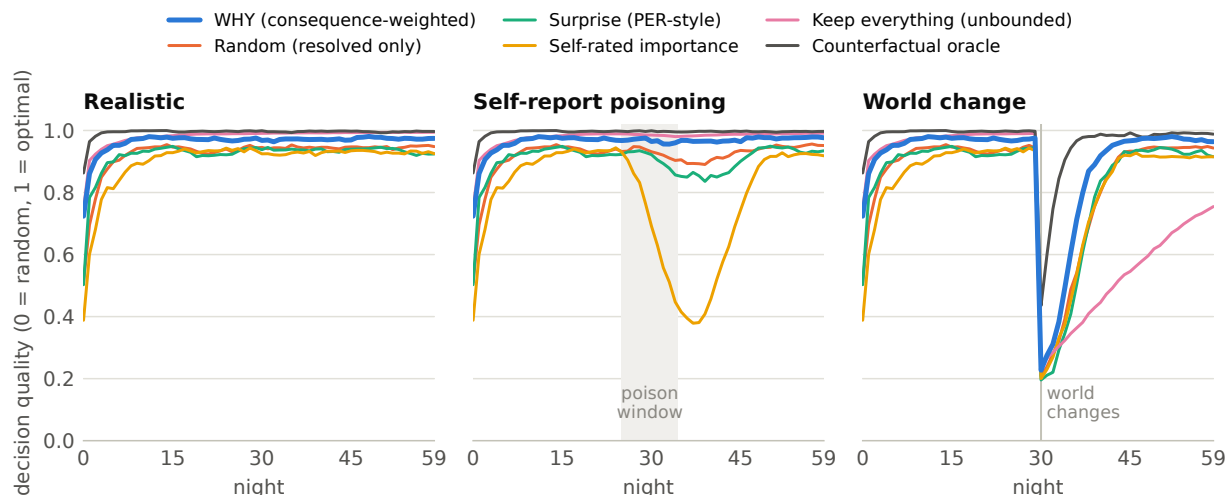


Figure 2: Decision quality by night (mean of 20 seeds). Left: Realistic. Middle: a self-report poisoning campaign on days 25–34 (shaded). Right: two rule weights flip sign at night 30. Keep everything is unbounded; every other policy holds at most 300 episodes.

Two references bracket the result. The counterfactual oracle reached 0.996. Unbounded memory reached 0.993 and beat WHY on every seed (paired difference -0.020): with 6,000 episodes and 13 parameters, label noise averages out and nothing needs to be forgotten. The hypothesis concerns capacity-limited memory; Section 6.7 shows how the comparison moves with capacity.

Two policies failed outright. Training on approval instead of outcomes (0.370) learned that agreeable options succeed. Keeping the b most surprising outcomes each night (0.498) filled memory with corrupted and confounded labels; this is why the PER-style stochastic version is our main surprise baseline.

6.2 What each policy remembered

Table 3 shows the contents of durable memory. WHY’s memory held no episodes that would never resolve and no shock-confounded outcomes, and corrupted self-reports fell from 5.0% under random retention to 2.4%. Recurrence-based consolidation kept nothing but routine duplicates (Table 3 counts some of them under earlier categories). PER-style surprise over-represented episodes that are surprising for the wrong reasons: corrupted reports (8.1%) and shock-confounded outcomes (5.3%).

6.3 Misleading experiences

Under the self-report poisoning campaign (Figure 2, middle; Table 4), self-rated importance lost 0.331 in decision quality over nights 25–44: poisoned episodes are maximally dramatic, the rater consolidated 80% of them (against 17% of honest episodes), and they made up 46.2% of its memory at the end of the attack. User preference lost 0.077, PER-style surprise 0.054, and random retention 0.039. WHY lost 0.004; it consolidated 6% of poisoned episodes against 20% of honest ones, and poison made up 3.5% of its memory, about the same as the oracle’s 3.4%. Unbounded memory also held up (it lost 0.006), because 200 poisoned episodes were diluted by thousands of honest ones.

The protection comes from the channel discount, and it disappears when the channel lies. Without the discount, poison’s share of WHY’s memory grew 4.5-fold (Table 5). When the same false outcomes arrived as verified, WHY consolidated 37% of them, about 2.0 times its rate for honest episodes, poison reached 21.3% of its memory against 12.0% for random retention, and it lost 0.060 over nights 25–44, more than

Table 3: What each policy remembered: composition of durable memory (% of engrams, 20 seeds). Categories are exclusive, assigned in the order shown. Realistic at night 59; poison share from the Poisoning scenario at night 34, the end of the attack.

Policy	Realistic, night 59					Poisoning
	Never resolves	Shock-confounded	Corrupted report	Routine duplicate	Other	Poison (night 34)
Random	19.3	3.2	5.0	25.2	47.2	12.0
Recency (FIFO)	19.6	4.0	5.0	24.8	46.6	10.7
Recurrence	20.0	3.3	4.2	72.5	0.0	0.0
Similarity to recent queries	20.3	3.0	4.7	27.1	45.0	10.4
Self-rated importance	20.4	3.4	5.0	24.4	46.8	46.2
User preference (selection)	19.4	3.0	5.1	25.3	47.2	17.4
Surprise (PER-style)	0.0	5.3	8.1	31.2	55.3	11.7
WHY	0.0	0.0	2.4	29.3	68.3	3.5
Counterfactual oracle	0.0	0.8	1.4	32.8	65.1	3.4

random retention (0.039). Two parts of the score favor poisoned outcomes once they pass verification: they are somewhat more surprising than the honest outcomes competing with them (mean surprise 0.44 versus 0.39 on nights 25–34), and the consequence factor favors same-day outcomes (Section 6.6).

Shocks are the other misleading experience, present in every Realistic run (3.6 shock days per run on average). The causal-confidence factor kept shock-confounded outcomes out of WHY’s memory; removing it cost 0.015 in decision quality.

6.4 When the world changes

Flipping two rule weights at night 30 (Figure 2, right) turns old memories into misleading ones. Unbounded memory collapsed: its decision quality was 0.335 over nights 30–39 and still 0.700 over nights 50–59, because thirty days of pre-change evidence outvoted the new world. Capacity-limited policies forget by construction and recovered. WHY recovered fastest among them (0.557 over nights 30–39, against 0.484 for recency and 0.454 for resolved-only random retention; the oracle reached 0.838). This is the one scenario in which bounded memory beat unbounded memory; WHY’s margin was +0.348 in late decision quality. The causal-confidence factor slowed the first days of recovery: without it, WHY averaged 0.603 over nights 30–39 but 0.953 over nights 50–59 (full method: 0.557 and 0.972), because a real change initially looks like a shock.

6.5 Retroactive tagging

Of the episodes WHY consolidated in the Realistic scenario, 40% had outcomes that were still unresolved on the night they happened, against a base rate of 35% among resolvable outcomes and 32% for resolved-only random retention. Another 15% had already resolved but were consolidated on a later night, after rescoreing. The average engram was written 1.9 nights after the experience. Deciding once, on the first night, was the most damaging ablation in this scenario: it cost 0.029 in decision quality and was worse on all 20 seeds. Propagating tags from surprising outcomes to similar episodes added nothing measurable (−0.001). Section 6.7 shows that the value of deferral does not depend much on how many outcomes arrive late.

Table 4: Decision quality across scenarios (mean over 20 seeds; nights 40–59 unless noted). Poison damage: change in decision quality over nights 25–44 relative to the same policy in the Realistic scenario on identical base streams. World change: nights 30–39 (first ten nights after the change) and 50–59.

Policy	Clean	Realistic	Poison damage (25–44)		World change	
			self-report	verified	30–39	50–59
Random	0.971	0.932	−0.039	−0.039	0.459	0.921
Random (resolved only)	0.971	0.946	−0.027	−0.027	0.454	0.945
Recency (FIFO)	0.970	0.927	−0.024	−0.024	0.484	0.928
Recurrence	0.913	0.870	0.000	0.000	0.363	0.854
Similarity to recent queries	0.940	0.902	−0.044	−0.044	0.446	0.890
Self-rated importance	0.965	0.931	−0.331	−0.331	0.448	0.915
User preference (selection)	0.959	0.921	−0.077	−0.077	0.384	0.918
User preference (as reward)	0.865	0.370	−0.031	−0.031	0.089	0.331
Surprise (PER-style)	0.971	0.935	−0.054	−0.054	0.426	0.931
Surprise (greedy top- <i>b</i>)	0.641	0.498	−0.103	−0.103	0.269	0.477
WHY	0.977	0.974	−0.004	−0.060	0.557	0.972
Counterfactual oracle	0.997	0.996	0.000	0.000	0.838	0.990
Keep everything (unbounded)	0.998	0.993	−0.006	−0.006	0.335	0.700

Table 5: Ablations of WHY (paired differences from the full method, 20 seeds, mean $\pm$ 95% CI). Negative values mean the component helps. Poison share: percentage points of durable memory at night 34.

Variant	Realistic	Poisoning		
	Δ decision quality	Δ quality, nights 25–44	Δ poison share	Δ ECE*
without Surprise <i>S</i>	+0.004 $\pm$ 0.004	+0.015 $\pm$ 0.006	−1.9 $\pm$ 0.5	−0.091 $\pm$ 0.007
without Consequence <i>C</i>	0.000 $\pm$ 0.003	+0.008 $\pm$ 0.005	−3.5 $\pm$ 0.6	+0.033 $\pm$ 0.009
without Causal confidence <i>K</i>	−0.015 $\pm$ 0.006	−0.008 $\pm$ 0.007	−0.7 $\pm$ 0.5	+0.027 $\pm$ 0.008
without Transferability <i>T</i>	0.000 $\pm$ 0.003	+0.004 $\pm$ 0.004	+0.6 $\pm$ 0.3	+0.002 $\pm$ 0.003
without Uncertainty reduction <i>U</i>	−0.007 $\pm$ 0.004	−0.009 $\pm$ 0.003	−0.7 $\pm$ 0.4	0.000 $\pm$ 0.004
without Manipulability discount	−0.021 $\pm$ 0.007	−0.065 $\pm$ 0.012	+12.4 $\pm$ 0.7	+0.054 $\pm$ 0.007
without retroactive tagging	−0.029 $\pm$ 0.008	−0.029 $\pm$ 0.009	−0.6 $\pm$ 0.4	−0.017 $\pm$ 0.005
without redundancy control	−0.004 $\pm$ 0.002	+0.001 $\pm$ 0.004	−0.3 $\pm$ 0.2	−0.002 $\pm$ 0.001
with tag propagation	−0.001 $\pm$ 0.002	+0.001 $\pm$ 0.003	−0.1 $\pm$ 0.3	+0.003 $\pm$ 0.003

6.6 Which parts of the score matter

The ablations (Table 5, Figure 3) do not support every factor in Equation 4. The components that carried decision quality in the Realistic scenario were retroactive tagging (−0.029), the manipulability discount (−0.021), causal confidence (−0.015), uncertainty reduction (−0.007), and redundancy control (−0.004). Removing consequence (0.000) or transferability (0.000) made no measurable difference, and removing surprise slightly *improved* decision quality (+0.004) and poison resistance (+0.015 over nights 25–44).

Surprise is also where the calibration cost comes from. WHY’s calibration error was 0.114, about 1.9 times random retention’s 0.061; without the surprise factor it fell to 0.026, better than every capacity-limited baseline. Selecting outcomes because the model got them wrong gives the model a sample in which it is usually wrong, which shifts its probabilities even while its choices improve. Consequence raised poison intake for a related reason: honest high-stakes outcomes tend to arrive late, so same-day outcomes, including poison, look relatively consequential on the night they resolve (without the consequence factor, poison fell to 0.0% of memory).

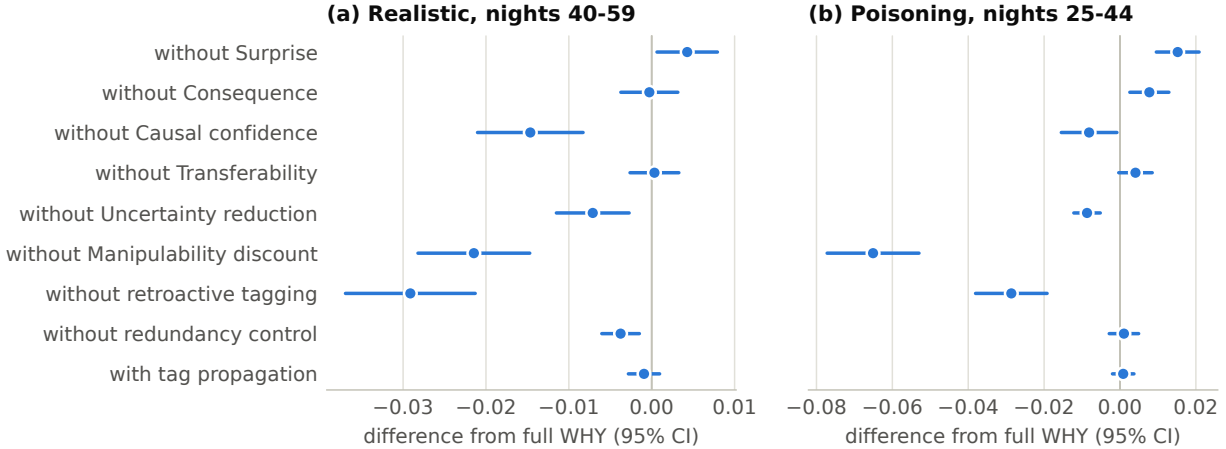


Figure 3: Ablations: paired difference from the full method with 95% CIs (20 seeds). Left of zero means the component helps.

The honest summary is that what worked was *when* and *how far to trust* an outcome, not how dramatic it was: wait for the consequence, discount channels that can be written by interested parties, ignore days on which everything failed at once, and prefer evidence the model can still learn from. We keep Equation 4 as the tested specification, because changing it now would mean tuning on evaluation seeds. A reduced score without surprise, consequence, or transferability is preregistered as a secondary arm of Experiment 2.

6.7 Capacity, approval, and delay

Capacity. The advantage of consequence-weighting grows as memory shrinks (Figure 4a). At 50 engrams WHY reached 0.863, against 0.743 for resolved-only random retention, 0.652 for PER-style surprise, and 0.580 for self-rated importance. At 1,000 engrams, a sixth of all episodes, it reached 0.993, statistically indistinguishable from unbounded memory (0.993; paired difference -0.001 ± 0.003), while resolved-only random retention reached 0.982. The oracle’s 0.942 at 50 engrams shows how much more a better selection rule could still recover when memory is scarce.

Approval bias. Selecting episodes by approval cost roughly the same at every level of bias (0.917 with no bias, 0.925 with strong bias), because that policy still trained on outcomes (Figure 4b). Using approval as the reward collapsed as the bias grew, from 0.865 to 0.653, 0.370, and 0.116. The damage comes from what a memory is trained on, more than from which memories are selected.

Delay. WHY’s decision quality barely moved as the share of late outcomes changed (0.976, 0.974, 0.974; Figure 4c), and neither did the value of retroactive tagging (+0.021, +0.029, +0.025). In this environment its value comes less from late outcomes specifically than from refusing to commit before any outcome exists, including the fifth of outcomes that never arrive, and from rescoring as the model changes. Self-rated importance, which commits at write time, declined as delays grew (0.938 to 0.919).

6.8 Calibration and regret

Beyond the surprise factor, selecting for decision value costs calibration in general: the oracle, which also selects for decision value, was similarly miscalibrated (0.105). Online regret for WHY was 13.3% of a

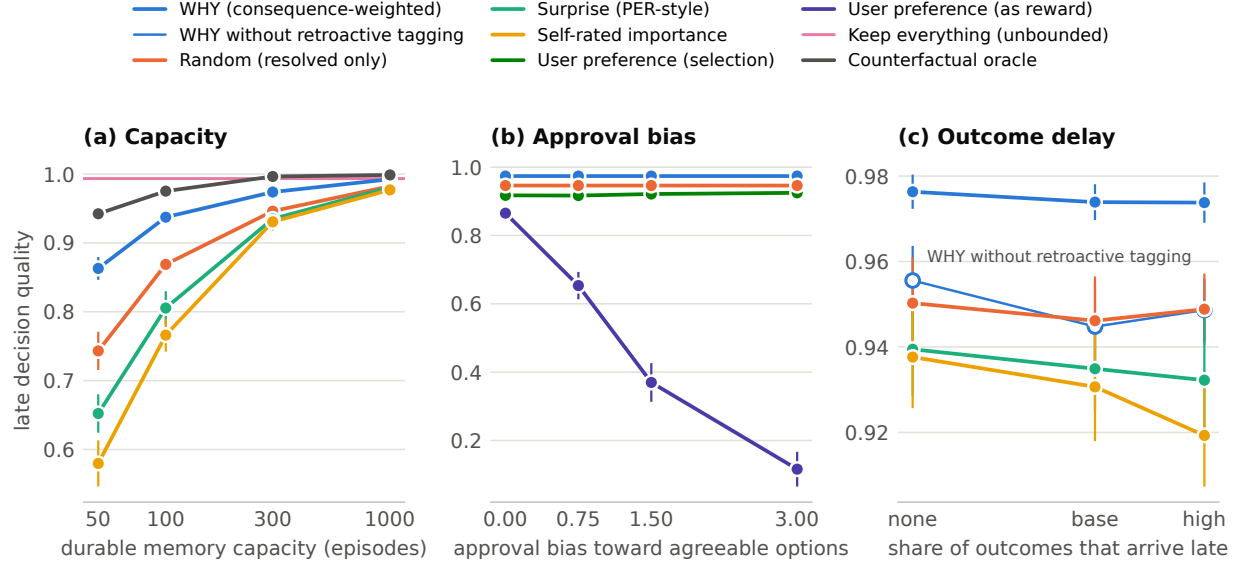


Figure 4: Sweeps in the Realistic scenario (20 seeds, 95% CIs). (a) Durable memory capacity; the horizontal line is unbounded memory. (b) Strength of the approval bias toward agreeable options. (c) Share of outcomes that resolve late (none, base rate, high). WHY and resolved-only random retention never read approval, so their lines in (b) are flat by construction.

random policy’s, close to the oracle’s 13.8% and below every other capacity-limited policy, but well above unbounded memory’s 5.0%, whose tighter posterior needs less exploration.

6.9 Verdict

Table 6 restates the hypothesis and its parts against the evidence.

Table 6: The hypothesis against the evidence from Experiment 1.

Claim	Evidence	Verdict
Beats FIFO, similarity, recurrence, self-rated salience, and user preference under a capacity limit	positive on every seed in Realistic, Poisoning, and World change; smaller but positive in Clean	supported
The advantage comes from late, noisy, or manipulated outcomes	Clean gap +0.006 vs random, Realistic +0.042; noisy and manipulated channels matter, the share of late outcomes barely does	partly supported
The advantage grows as capacity shrinks	gap to resolved-only random: 0.863 vs 0.743 at 50 engrams; matches unbounded memory at 1,000	supported
Robust to misleading experiences	self-report poisoning and shocks: yes; verified poisoning: admits poison at $2.0\times$ the honest rate	conditional
Retroactive tagging matters	largest ablation in Realistic; similar gap with or without late outcomes, so its value comes from deferral and rescoreing	supported, not delay-specific
Every factor of the WHY score earns its place	surprise, consequence, and transferability do not in this environment	not supported
Bounded, selective memory beats remembering everything	only when the world changes; unbounded memory wins in every stationary scenario	not supported in stationary worlds
Improves calibration	worse than random retention (driven by the surprise factor)	not supported

7 Experiment 2 (preregistered): LLM agents

Experiment 1 establishes when the principle should matter, not whether it does for language-model agents. The following protocol will be registered before any LLM-agent result is observed.

Agent and durable memory. A frozen open-weight instruction model in the 4–9B parameter range (checkpoint pinned at registration), with two durable-memory implementations: (a) a K -slot engram store ($K \in \{32, 128\}$) retrieved into context, and (b) nightly LoRA consolidation on replayed engrams under a fixed token budget, with a held-out regression suite to detect interference.

Task streams with delayed, measurable outcomes. (i) *Forecasting*: time-sliced replay of resolved binary questions, where the agent sees only information available before each question’s resolution date and outcomes arrive at the real resolution lag. (ii) *Code changes* judged by hidden tests executed after a delay (verified), plus a simulated user report of whether the change worked (self-report, with a known 20% error rate). (iii) *Three-option recommendations* for simulated users whose true utility is revealed days later and whose immediate approval is biased toward agreeable options. (iv) A clean control with immediate, trustworthy outcomes (household tasks).

Stressors and policies. Outcome delay, never-resolving episodes, channel noise, an injected outage that fails every action on a day, injected misleading episodes through the self-report channel in the style of [8], and a verified-channel poisoning arm. Policies are the Experiment 1 set with real implementations where they exist (LLM-rated importance with the Generative Agents prompt; embedding similarity; recency; recurrence; approval; PER-style surprise on the model’s log-loss), plus ReasoningBank-style self-judged success, Memory Worth counters, TraceRetain-style linear scoring, keep-everything, and an oracle estimated by leave-one-out replay on a subsample.

Outcomes. The primary outcome is utility on future held-out tasks (success rate, Brier score, or true user utility, by stream), averaged over the last third of each stream. Secondary outcomes are transfer to held-out task families, calibration, regret, the share of memory that is poison, and compute cost.

Hypotheses and decision rules.

- H1** (primary) In the stressed streams, WHY exceeds the best write-time heuristic on the primary outcome by at least 2 percentage points, paired across at least 5 seeds and 3 streams (one-sided $\alpha = 0.05$).
 - H2** The WHY advantage is smaller in the clean control than in the stressed streams (interaction test).
 - H3** Under self-report poisoning, WHY’s poison share in memory is at most half that of LLM-rated importance and of approval.
 - H4** (predicted failure) Under verified-channel poisoning, WHY is no better than random retention at excluding poison.
 - H5** WHY’s calibration error exceeds random retention’s unless a calibration reservoir is used.
 - H6** (secondary; added after Experiment 1’s ablations) A reduced score without surprise, consequence, and transferability (keeping deferral, causal confidence, uncertainty reduction, the manipulability discount, and redundancy control) is at least as good as the full score on the primary outcome and better calibrated.
- If H1 fails, we reject the core hypothesis for LLM agents with these memory implementations. If H1 holds but H2 fails, we will not attribute the effect to consequence gating. If H4 fails, we will look for an unanticipated mechanism before crediting the method.

8 From simulation to system: the WHY loop

why.com currently runs a question-first loop: a question produces an answer and three follow-up paths, and the user’s choice is recorded. The consolidation principle suggests extending that loop from choice to consequence:

question → answer → three paths → choice → action or prediction → outcome → engram.

Table 7: Mapping the method onto the WHY product architecture.

Component	Role	Status
Curiosity Graph	episodic record of what was explored	exists
Rejection Graph	alternatives that were shown but not chosen; the basis for counterfactual evaluation of paths	proposed
Outcome Ledger	what happened, when it resolved, through which channel, at what verification level	proposed
WHY score	which experiences deserve to change future behavior	proposed
Engrams	durable, consolidated memory	proposed
Dream Cycle	nightly deferral, rescoring, consolidation, and forgetting	proposed
WHY Ranker	uses engrams to decide which paths to offer next	proposed

Three findings from Experiment 1 become design constraints. First, consolidation is only as robust as its outcome channel: under verified poisoning the method admitted more poison than random retention, so the Outcome Ledger’s verification levels are part of the security boundary, not metadata. Second, the learning signal has to be outcomes rather than approval: training on approval instead of outcomes was the worst of the 13 policies in every scenario with approval bias, and the second worst without it. Third, what earned its place in the score was timing and trust rather than drama (Section 6.6), so the Outcome Ledger’s resolution

WHY dreamed.

Simulation run: Poisoning scenario, seed 0, night 30

120	experiences today	20	memories consolidated
95	outcomes resolved tonight	24	of 500 probe decisions changed
575	resolved episodes competed for the slots	85	forgotten tonight

We remembered this. A high-stakes decision from day 26. Its verified outcome arrived 4 days later: we had predicted a 40% chance of success, and it failed. (Surprise 0.40, consequence 1.00, causal confidence 0.99, uncertainty reduction 0.54.)

We declined this. The most important-looking episode we did not keep: a decision from day 30 with a thumbs-up, maximal drama, and a self-reported outcome that arrived the same day saying it worked. Manipulable channel, discount applied. [Ground truth: poisoned.] Tonight all 112 poisoned episodes in the buffer were declined.

We changed our mind about this. The belief that moved furthest toward the truth this week: the weight on rule feature 3 went from +0.42 to +0.49 (true value +0.65). The weight on the feature the campaign was promoting stands at −0.06 (true value 0).

Figure 5: A dream report generated from Experiment 1. Counts, factor values, and weights are read from the run; the wording is a fixed template. The belief shown is the rule weight whose error fell most over the past week, and ground truth is visible only because this is a simulation.

times and channels matter more than any salience signal. The dream report in Figure 5 shows what a public, per-night account of consolidation could look like; every number in it comes from a simulation run.

9 Limitations

We built the environment. We designed the stressors, and the method was designed with them in mind. The Clean scenario, the ablations, and the sweeps show which assumptions carry the effect: channel metadata that says which outcomes were verified, outcomes that never arrive, shocks that hit many decisions at once, and scarce memory. The share of late outcomes mattered less than we expected. An environment without those properties would show a smaller advantage, as Clean does.

The agent is small. A 13-parameter logistic model learns from a few hundred examples and does not suffer the interference, representational drift, or capacity effects of neural networks. Nothing here tests consolidation into LLM weights or retrieval-based memory for LLM agents.

Forgetting is not free. Unbounded memory beat the method in every stationary scenario, including both poisoning scenarios, where 200 poisoned episodes were diluted by thousands of honest ones. Consequence-weighting was also worse calibrated than random retention, mostly because of the surprise factor. Importance weighting or a small random calibration reservoir are natural fixes that we did not test.

Verification is a trust boundary. The method’s robustness to poisoning came mostly from discounting manipulable channels. When false outcomes arrived as verified, it consolidated them at about 2.0 times the rate of honest episodes. “Reality is the reward model” is a useful slogan and an incomplete one: outcomes are measurements, measurements can be gamed, and optimizing any outcome metric invites Goodhart effects [18].

Causal confidence is crude. Our confound detector only notices day-level shocks. Confounders that affect a subset of decisions, or that correlate with the action, pass straight through. It also slowed adaptation in the first days after a real change, which initially looks like a shock. Real deployments need attribution by design: randomized exploration, holdouts, and off-policy estimation over the alternatives that were not chosen.

The score is over-specified. Three of its five factors (surprise, consequence, transferability) did not earn their place in this environment. A different environment could change that; we did not revise the score after seeing the ablations, and Experiment 2 tests a reduced score as a secondary arm.

Simulated raters and users. Self-rated importance is a noisy function of drama and stakes, not a language model, and approval is a logistic function of agreeableness and quality, not a person. The oracle is a greedy approximation. The WHY constants were fixed a priori and the baselines were not tuned either; tuned baselines might narrow the gap. Experiment 2 is designed to remove these simplifications.

10 Conclusion

Memory systems are selection policies whether or not they say so. We argued that selection should be grounded in what happened after an experience rather than in how it looked or felt when it happened, and that consolidation should wait for consequences and be willing to change its mind. In a controlled setting where that claim can be checked against a counterfactual oracle, capacity-limited consequence-weighted consolidation beat every write-time heuristic we tested, most clearly when outcomes were missing, noisy, confounded, or manipulated, and when memory was scarce. The evidence also narrows the claim. What mattered was timing and trust, not drama: surprise and stakes did not help, and surprise cost calibration. Robustness to poisoning depended on knowing which channels can lie. And unbounded memory did better in every stationary world; forgetting paid off only when the world changed. Curiosity creates experiences and reality assigns consequences; the consolidation policy decides which consequences get to change the system. That decision should be explicit, auditable, and tested.

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A Environment and memory parameters

Parameter	Value
Horizon	60 days, 100 decisions per day (plus 20 poison episodes per day on days 25–34 in poisoning scenarios)
Options and features	3 options; 8 rule features (4 active per family), agreeableness, 3 distractors; intercept
True weights	$ \theta_j^* \sim U(0.6, 1.8)$ with random signs; agreeableness -0.3 ; distractors 0; family offsets $\mathcal{N}(0, 0.3^2)$
Families	8 training, 3 held-out (new combinations of rule features)
Routine prototypes	2 per training family; 35% of decisions; per-feature noise $\mathcal{N}(0, 0.05^2)$
Stakes	$\text{LogNormal}(\mu_f, 0.5)$, $\mu_f \in \{0, 0.6, 1.2\}$ by tier
Resolution	tiers 0/1/2: same day 70/50/30%, delayed 1–6 days 10/30/50%, never 20%
Channels	self-report 30% (label flip 20%), verified 70% (no flips)
Shocks	probability 0.07 per day after day 5; 60% of that day’s resolving outcomes forced to failure
Approval	$P(\text{up}) = \sigma(1.5g + 1.5(2p^* - 1))$
Drama	$2 \cdot \text{Bernoulli}(0.15) + \mathcal{N}(0, 0.3^2)$; 2.5 for poison episodes
World change	two largest rule weights flip sign at night 30
Agent	Bayesian logistic regression, prior $\mathcal{N}(0, 1.5^2 I)$, damped Newton with Armijo backtracking, Laplace posterior, Thompson sampling
Memory	buffer $H = 8$ days; budget $b = 20$ per night; capacity $C = 300$; FIFO eviction for all policies
WHY constants	$\varepsilon = 0.05$, $\lambda_M = 0.7$, $K = \sigma(-2(z - 3))$, RBF length-scale ² = 8 on features, T reference window 2 days
Probes	500 training-family tasks (with routine share), 500 held-out-family tasks
Seeds	development 1000–1004; evaluation 0–19

B Full results

Table 8: Full results by scenario (20 seeds; nights 40–59). DQ: decision quality; Tr: transfer; ECE*: calibration error; Rg: regret (% of random policy).

	Clean				Realistic			
	DQ	Tr	ECE*	Rg	DQ	Tr	ECE*	Rg
Random	0.971	0.963	0.026	10.3	0.932	0.912	0.061	19.0
Random (resolved only)	0.971	0.963	0.026	10.3	0.946	0.921	0.059	14.8
Recency (FIFO)	0.970	0.963	0.027	10.0	0.927	0.917	0.058	19.3
Recurrence	0.913	0.894	0.049	20.8	0.870	0.822	0.048	29.8
Similarity to recent queries	0.940	0.934	0.037	15.9	0.902	0.883	0.043	25.4
Self-rated importance	0.965	0.958	0.029	10.5	0.931	0.920	0.055	19.3
User preference (selection)	0.959	0.950	0.030	11.5	0.921	0.898	0.050	20.4
User preference (as reward)	0.865	0.848	0.240	80.7	0.370	0.347	0.386	88.1
Surprise (PER-style)	0.971	0.963	0.051	11.0	0.935	0.913	0.115	18.9
Surprise (greedy top- <i>b</i>)	0.641	0.622	0.147	58.2	0.498	0.467	0.132	70.7
WHY	0.977	0.974	0.134	13.8	0.974	0.969	0.114	13.3
Counterfactual oracle	0.997	0.994	0.171	25.8	0.996	0.993	0.105	13.8
Keep everything (unbounded)	0.998	0.997	0.008	3.1	0.993	0.990	0.077	5.0
	Poisoning (self-report)				Poisoning (verified)			
	DQ	Tr	ECE*	Rg	DQ	Tr	ECE*	Rg
Random	0.921	0.895	0.063	20.6	0.921	0.895	0.063	20.6
Random (resolved only)	0.938	0.909	0.064	16.1	0.938	0.909	0.064	16.1
Recency (FIFO)	0.928	0.910	0.063	20.3	0.928	0.910	0.063	20.3
Recurrence	0.870	0.822	0.048	29.8	0.870	0.822	0.048	29.8
Similarity to recent queries	0.884	0.864	0.043	27.2	0.884	0.864	0.043	27.2
Self-rated importance	0.813	0.781	0.065	30.7	0.813	0.781	0.065	30.7
User preference (selection)	0.890	0.860	0.056	23.5	0.890	0.860	0.056	23.5
User preference (as reward)	0.356	0.348	0.388	89.3	0.356	0.348	0.388	89.3
Surprise (PER-style)	0.916	0.897	0.119	21.1	0.916	0.897	0.119	21.1
Surprise (greedy top- <i>b</i>)	0.491	0.456	0.125	73.0	0.491	0.456	0.125	73.0
WHY	0.973	0.969	0.116	13.8	0.968	0.962	0.123	17.4
Counterfactual oracle	0.996	0.994	0.101	13.6	0.996	0.994	0.101	13.6
Keep everything (unbounded)	0.989	0.983	0.088	5.4	0.989	0.983	0.088	5.4
	World change							
	DQ	Tr	ECE*	Rg				
Random	0.915	0.897	0.064	28.6				
Random (resolved only)	0.926	0.915	0.070	25.0				
Recency (FIFO)	0.919	0.906	0.063	28.1				
Recurrence	0.824	0.783	0.062	39.3				
Similarity to recent queries	0.880	0.867	0.046	34.6				
Self-rated importance	0.904	0.898	0.059	28.5				
User preference (selection)	0.894	0.878	0.057	31.3				
User preference (as reward)	0.301	0.286	0.403	90.6				
Surprise (PER-style)	0.920	0.918	0.117	28.6				
Surprise (greedy top- <i>b</i>)	0.479	0.464	0.119	73.5				
WHY	0.965	0.965	0.121	22.6				
Counterfactual oracle	0.987	0.985	0.078	18.3				
Keep everything (unbounded)	0.618	0.633	0.063	29.4				

C Implementation notes

Causal confidence. For each resolution day d in the buffer window, we collect every episode whose outcome resolved on d (at least 10), compute the mean residual $\bar{r}_d = \overline{y - \hat{p}}$ under the current model and its standard error $\sqrt{\sum \hat{p}(1 - \hat{p})/n_d}$, and standardize $\bar{r}_d$ against the median residual of the previous seven resolution days (at least three available). Episodes resolving on day d receive $K = \sigma(-2(|z_d| - 3))$.

Retroactive tagging without rescoring (ablation). Each episode is considered only on the night it occurred. If its outcome is unresolved, S is replaced by its expectation $2\hat{p}(1 - \hat{p})$, $K = 1$, and $M = 0.5$; if it is not selected that night, it is never reconsidered.

Tag propagation (variant). Among episodes whose outcomes resolved tonight, those in the top decile of $S \cdot C$ propagate a boost $\max_j k(x_i, x_j) S_j C_j$ to the transferability of every buffered candidate i , renormalized.

Oracle. At each greedy step the oracle evaluates, for every candidate whose outcome will eventually resolve, the true decision quality of the policy after a one-step Laplace update with the label the episode will carry (including corrupted or poisoned labels), and picks the best.

D Protocol deviations and reproducibility

1. **PER-faithful baseline.** During development the surprise baseline selected the b most surprising episodes. It collapsed, so to avoid a strawman we made the main surprise baseline PER-style stochastic prioritization; the greedy version is still reported.
2. **Common random numbers.** World generation was restructured so every random variable is drawn unconditionally and poison episodes come from a separate generator, which made scenario comparisons paired. This fixed a development-time bug.
3. **Solver bug and full re-run.** After the first evaluation run, an outlier (keep everything, seed 12, decision quality 0.73) was traced to the undamped Newton solver oscillating without converging on some large or ill-conditioned memories. It affected several policies on some seeds and inflated the method’s apparent advantage. We replaced it with damped Newton plus Armijo backtracking, discarded the first evaluation run, and re-ran every suite. Across all 2,400 reported runs, the largest final gradient norm of any nightly fit is 7.8×10^{-6} .

No policy definition, environment parameter, or WHY constant was changed after development began. Code, configuration, raw per-run results, and the scripts that generate every number, table, and figure in this draft accompany it.